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Objective:	To convert a measurement from a unit to another unit.
Measurement:	Mass
1st Given Unit:	Customary unit (Ounce)
To Convert to:	Metric unit (Gram)
2nd Given Unit:	Metric unit (Gram)
To Convert to:	Customary unit (Ounce)
Container Used:	Peanut Butter Jar (<i>Please see bottom center of container</i>)



Convert 28 oz to g.

From Given Tables:

$$16 \text{ oz} = 1 \text{ lb}$$

$$453.59237 \text{ g} = 1 \text{ lb}$$

$$453.59237 \text{ g} = 16 \text{ oz}$$

We shall use the proportional reasoning method to convert 28 oz to g.

Proportional reasoning method.

Let x = mass of 28 oz in g

$$\frac{x}{28} = \frac{453.59237}{16}$$

Multiply both sides by 28

$$28 * \frac{x}{28} = 28 * \frac{453.59237}{16}$$

$$x = \frac{28 * 453.59237}{16}$$

$$x = \frac{12700.58636}{16}$$

$$x = 793.7866475$$

$$x \approx 794 \text{ g} \dots \text{rounded to the nearest whole number}$$

$$28 \text{ oz} \approx 794 \text{ g}$$

This confirms the quantity in *g* labeled on the peanut butter jar.

Second conversion is to convert 794 *g* to *oz*.

From Given Tables:

$$1 \text{ lb} = 16 \text{ oz}$$

$$1 \text{ lb} = 453.59237 \text{ g}$$

$$453.59237 \text{ g} = 16 \text{ oz}$$

Unity Fraction Method

*Let x = mass of 794 *g* in *oz**

$$794 \text{ g} * \frac{...lb}{...g} * \frac{...oz}{...lb} = x$$

$$794 \text{ g} * \frac{1 \text{ lb}}{453.59237 \text{ g}} * \frac{16 \text{ oz}}{1 \text{ lb}} = x$$

$$x = \frac{794 * 1 * 16}{453.59237 * 1}$$

$$x = \frac{12704}{453.59237}$$

$$x = 28.00752578796685$$

$$x \approx 28 \text{ oz ...rounded to the nearest whole number}$$

This confirms the quantity in *oz* labeled on the peanut butter jar.